



GROMS: GPS-free real-time UAV orthophoto mapping via terrain constraint monocular SLAM

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ABSTRACT

To improve the coverage of individual Unmanned Aerial Vehicle (UAV) images and enable comprehensive environmental perception, it is essential to generate wide-field, large-scale, and metrically accurate orthophotos. However, traditional methods depend heavily on the Global Positioning System (GPS), exhibit high computational complexity, and are susceptible to drift and distortion in GPS-free environments. To overcome these challenges, we propose GROMS, a GPS-free, map-prior-assisted (DEM and satellite imagery) real-time UAV orthophoto mapping system. GROMS utilizes georeferenced satellite imagery and digital elevation model (DEM) solely as map priors to derive virtual control points (VCPs) and constrain monocular SLAM, without relying on any GNSS or IMU measurements. Specifically, we design an efficient keyframe selection and dynamic block matching module to enable robust alignment between UAV and satellite images. We further introduce a terrain-constrained monocular SLAM framework incorporating VCPs, enabling drift-free and high-precision pose estimation in the absence of GPS/IMU. Finally, a hybrid ortho-rectification and incremental multi-band blending strategy is proposed for seamless, real-time orthophoto generation. Extensive experiments in both typical and challenging scenarios validate the robustness, accuracy, efficiency, and visual fidelity of GROMS. The system achieves visual quality comparable to commercial software, with efficiency improvements of up to 7.6×. Under GPS-free conditions, GROMS attains mean absolute errors of 1.11 m and 0.64 m in the x- and y-directions, respectively, while sustaining stable end-to-end throughput with bounded latency ($p_{99} < 2.8$ s) during continuous online orthophoto generation. Demonstration videos of GROMS and the associated datasets are available at <https://github.com/YFS90/Real-time-orthophoto>.

1. Introduction

UAVs have recently seen broad use in diverse fields ranging from agriculture to military surveillance due to their low cost and operational flexibility (Nex et al., 2022; Zhang et al., 2023). However, the low flight altitude and narrow camera field of view (FoV) restrict each UAV image to a limited ground area. Real-time image stitching mitigates these limitations by generating Digital Orthophoto Maps (DOMs), which offer wide-area mosaics with high geometric precision and radiometric consistency. But DOM generation depends on accurate GPS data and intensive computation, making real-time mapping difficult, especially without GPS. This greatly limits their use in applications needing fast, accurate mapping.

A commonly used approach for rapid orthophoto generation is image stitching, which depends on feature matching and transformation models to align and merge images. Techniques like tie-point area ratio optimization (Kim et al., 2020), incremental stitching (Zhang et al.,

2020), and superpixel-based methods (Yuan et al., 2020) reduce distortion and seams, enabling real-time performance. However, assuming planar scenes makes them sensitive to uneven terrain, causing ghosting, misalignment, and distortion under changing light, significant parallax, or low-texture conditions. Structure-from-Motion (SfM) reconstructs 3D scenes from unordered multi-view images and estimates poses using incremental, global, or hierarchical strategies. Recent advances like fast georeferencing (Zhao et al., 2020b), real-time SfM-SLAM hybrids (Zhao et al., 2021), and multi-sensor fusion (He et al., 2025) have improved orthophoto efficiency and robustness. However, SfM is computationally intensive due to iterative bundle adjustment (BA) and high memory use, limiting real-time applicability. Additionally, they also rely on GPS or ground control points (GCPs), reducing effectiveness in unknown or GPS-free areas. Simultaneous Localization and Mapping (SLAM) based methods, considered online variants of SfM, enable real-time pose estimation and scene reconstruction, offering flexibility

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and low hardware requirements. Frameworks like Map2DFusion (Bu et al., 2016) and OpenREALM (Kern et al., 2020) use monocular SLAM and sensor fusion to incrementally create orthophotos. These methods are efficient and reasonably accurate. However, SLAM suffers from drift over long trajectories, especially in feature-sparse scenes, and struggles with low-overlap, high-resolution images (Zhao et al., 2020a). Moreover, monocular SLAM cannot estimate real-world scale without external references like GPS, limiting accuracy in GNSS-denied areas. Deep learning methods like Neural Radiance Fields (NeRF) and Gaussian Splatting show promise in generating True Digital Orthophoto Maps (TDOMs) by implicitly reconstructing scenes. Models like OrthoNeRF (Chen et al., 2024) and Tortho-Gaussian (Wang et al., 2024) handle distortion, occlusion, and complex geometry without DSMs or explicit 3D models. Despite strong performance in complex scenes, they need high-overlap images, accurate initial poses, and heavy computation. These constraints cause long training times and poor performance in large scenes, strong reflections, or thin structures, making them unsuitable for real-time or GPS-free use.

To enable accurate and efficient orthophoto generation, we propose GROMS, a GPS-free, map-prior-assisted (DEM and satellite imagery) real-time orthophoto mapping system. GROMS aligns UAV frames with tiled satellite imagery, estimates and optimizes UAV poses using virtual control points within a GPS/IMU-free SLAM framework, and performs real-time orthophoto generation and fusion using a hybrid orthorectification and local blending strategy. The main contributions of this work are as follows:

- (1) We developed GROMS, a real-time UAV orthophoto mapping system tailored for long-distance, large-scale, high-altitude, and multi-strip flights in GPS-free environments. Unlike conventional systems, GROMS eliminates the dependence on downward-facing imagery, GPS, and inertial measurement units (IMUs). Compared to commercial software, GROMS delivers comparable visual quality and accuracy while enhancing orthophoto generation efficiency by up to 7.6×.
- (2) To enable robust orthophoto generation in the absence of GPS and IMU, we introduce a novel UAV pose estimation and optimization approach based on terrain-constrained monocular SLAM. This approach extracts virtual control points to ensure drift-free and viewpoint-invariant performance, maintaining robustness under varying altitudes, complex flight trajectories, undulating terrain, low-texture environments, and seasonal appearance changes. In addition, a hybrid orthorectification and incremental multi-band blending strategy enables real-time orthophoto generation and fusion, mitigating color discrepancies, misalignments, and ghosting artifacts while enhancing both visual quality and geometric accuracy.
- (3) We perform extensive experiments on both public and self-collected datasets encompassing plains, hilly terrain, and urban as well as rural environments. Comparative evaluations with state-of-the-art methods and ablation studies demonstrate the robustness, accuracy, and efficiency of the GROMS system under diverse and challenging conditions.

The remainder of this manuscript is structured as follows: Section 2 reviews related work. Section 3 provides a detailed description of the proposed algorithm. Section 4 presents and discusses the experimental results. Section 5 concludes the study.

2. Related work

In recent years, extensive research has focused on orthophoto generation from aerial imagery. This section reviews recent advances in UAV orthophoto generation, focusing on stitching-based, SfM-based, SLAM-based, and deep learning-based methods.

2.1. Stitching methods

Stitching methods typically begin with image preprocessing, followed by feature point extraction and matching to establish correspondences. Subsequently, transformation models such as homography or affine transformations are estimated to align images within a unified coordinate system. For long-range multi-strip flights, Zhang et al. (2020) proposed a real-time incremental stitching framework that operates without GPS or GCPs. However, it remains challenged by non-planar scenes and color blending issues. Liu et al. (2017) proposed a model for medium- to high-altitude UAV images with low overlap. Although it achieves 0.758 m accuracy on real data, the method is restricted to planar scenes. Wan et al. (2020) proposed a stitching method that combines bundle adjustment with shape-preserving transformations. While effective in complex scenes, it performs poorly on linear features, requires manual tuning, and ignores 3D terrain. Li et al. (2023) developed a real-time incremental stitching framework incorporating a coarse-to-fine strategy. However, it is sensitive to low-altitude flights and non-planar terrain, often resulting in misalignments at image boundaries. SatellStitch (Wei et al., 2024) uses satellite imagery to aid UAV stitching without GCPs or GPS. It improves blending through adaptive ROI extraction and gradient-weighted cost maps, reducing drift and ghosting. Nonetheless, it remains limited to low-altitude and complex terrains. A study (Jin et al., 2023) presented a high-accuracy method for stitching low-overlap thermal infrared sweeping images. It achieves 0.753 m accuracy but struggles with complex terrain and relies on manual feature extraction.

2.2. SfM-based methods

SfM reconstructs 3D scenes by estimating camera poses and scene geometry from unordered images. According to reconstruction strategies, SfM methods can be categorized as incremental (Schonberger and Frahm, 2016), global (Moulon et al., 2013), or hierarchical (Farenzena et al., 2009; Y. Chen et al., 2020). Commercial 3D mapping software such as Photoscan (Agisoft, 2025) and Pix4DMapper (Pix4D, 2025) adopt hierarchical SfM frameworks as their core. SfM-based orthophoto generation typically involves feature detection, sparse reconstruction, bundle adjustment, dense point cloud creation, digital surface model (DSM) generation, and orthorectification. Study (Zhao et al., 2020b) proposed a fast georeferenced stitching method that integrates constrained matching, multihomography, GPS-aided rotation averaging, and planar-constrained global SfM to improve orthophoto generation efficiency. Zhao et al. (2021) developed RTSfM, a system combining SfM accuracy with SLAM's real-time performance. However, it relies heavily on high-precision GPS and degrades significantly under weak or absent GPS signals. He et al. (2025) introduced a fast orthophoto generation method based on multi-sensor fusion, enhancing traditional SfM by reducing computational complexity and mismatches.

2.3. SLAM-based methods

SLAM can be considered the online counterpart of SfM. SLAM recovers scene structure and camera poses in real time, offering better speed, lower hardware demands, and greater flexibility. Thus, many studies use SLAM for real-time aerial image mosaicking. Map2DFusion (Bu et al., 2016) utilized monocular SLAM to estimate poses and fit planes from sparse point clouds for incremental large-scale mosaicking. However, it depends on GPS for quality and performs poorly in non-planar scenes. Hinzmann et al. (2018) fused camera, IMU, and GPS data, using iSAM2 to optimize trajectories, reconstruct dense 3D point clouds, and generate orthophotos incrementally via back projection. TerrainFusion (Wang et al., 2019) employs monocular SLAM for real-time DSM reconstruction, generating local DEMs and orthophotos from sparse clouds and integrating them into global maps via tiled multiband fusion. However, it uses only sparse clouds, resulting in low-resolution

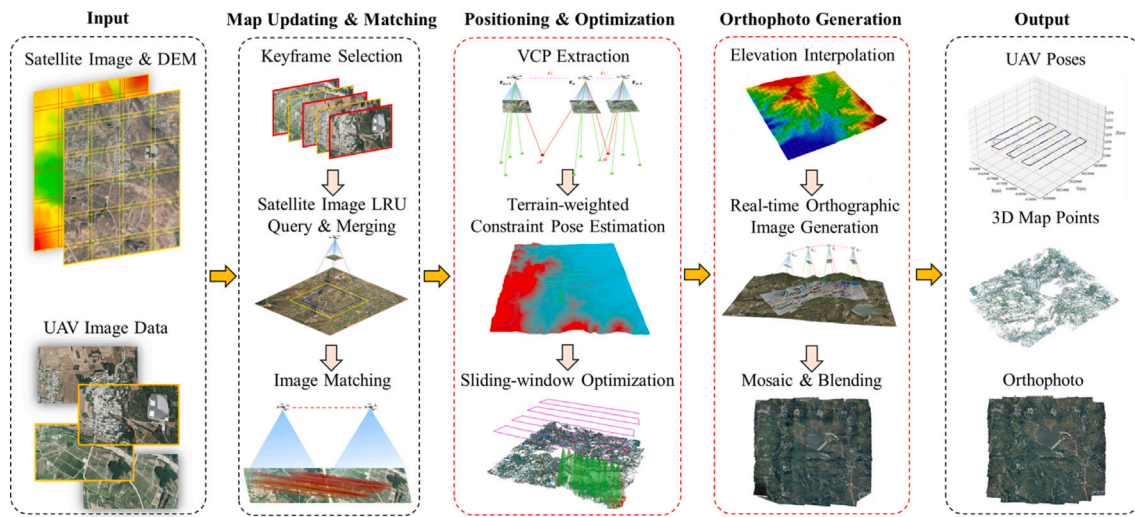


Fig. 1. The pipeline of GROMS. The system consists of three main modules: satellite map updating and image matching, UAV image pose estimation and optimization, and orthophoto generation and fusion.

outputs. DenseFusion (L. Chen et al., 2020) uses virtual stereo pairs and ELAS matching for dense depth estimation, with multi-frame filtering and weighted multiband rendering for high-quality fusion. However, GPS is needed for joint optimization and accurate georeferencing. Zhao et al. (2020a) developed a mobile visual SLAM-based method for real-time orthophoto of low-overlap aerial images. Although effective for mobile aerial imaging, it struggles with uneven terrain and moving targets. OpenREALM (Kern et al., 2020) integrates visual SLAM, dense reconstruction, and GPS for accurate pose estimation and surface modeling. Although grid-based orthorectification and probabilistic fusion produce high-quality orthophotos, the method's performance degrades in areas with sparse features. GeoROS (Gao et al., 2022) uses GPS and visual SLAM for real-time UAV mosaicking with accurate pose estimation and orthogonality-preserving fusion. It achieves 2.1 m GCP error and fast runtime but is less robust in difficult environments or with low-overlap imagery.

2.4. Deep learning and differentiable rendering-based methods

In recent years, deep learning has demonstrated superior performance in critical stages of orthophoto generation, including feature matching, camera pose estimation, and depth estimation. As a result, interest has grown in UAV-based TDOM generation using NeRF and Gaussian Splatting. Ortho-NeRF (Chen et al., 2024) and NeRF-Ortho (Yue et al., 2025) generate UAV TDOMs by tiling scenes, then implicitly reconstructing and rendering each tile using true-ortho-volume rendering. They outperform traditional approaches in complex scenes by minimizing distortions and effectively managing occlusions. However, they require high image overlap and long training time, which limits scalability and practical use in large-scale scenarios. Tortho-Gaussian (Wang et al., 2024) and Wang et al. (2025) adopt orthogonal splatting, a divide-and-conquer strategy, and geometry-aware depth estimation to produce high-quality, large-scale TDOMs. They surpass commercial tools in terms of edge sharpness, façade removal, and the handling of thin structures or low-texture regions while maintaining comparable accuracy. However, their scalability to city-scale environments, semantic interpretability, and robustness across diverse datasets need improvement.

In summary, although existing approaches have achieved notable progress, they continue to face critical limitations that restrict their applicability to real-world UAV orthophoto generation. Stitching-based methods are efficient and lightweight, but their strict planar-scene assumptions make them susceptible to ghosting and severe distortions in non-planar or large-scale environments. SfM-based pipelines

provide accurate 3D reconstruction but depend on GPS/GCPs and computationally intensive bundle adjustment, which prevents real-time deployment under GPS-free conditions. SLAM-based methods enable online mapping with modest hardware requirements, but they suffer from scale ambiguity, cumulative drift, and reduced robustness in low-texture or low-overlap imagery, particularly in the absence of GPS/IMU references. Deep learning-based approaches, such as NeRF and Gaussian Splatting, yield visually compelling results but require high image overlap, precise initialization, and long training times, thereby limiting scalability and preventing real-time use. These limitations highlight the necessity of developing GPS-free real-time orthophoto generation methods.

To address these challenges, we propose GROMS, a GPS- and IMU-free orthophoto mapping system that incorporates three key innovations: (1) a terrain-constrained monocular SLAM framework augmented with virtual control points (VCPs), which mitigates drift and enables scale-aware pose estimation; (2) a hybrid ortho-rectification and incremental multi-band blending strategy that reduces distortions, corrects color inconsistencies, and alleviates ghosting artifacts while improving geometric accuracy; (3) an efficient keyframe selection and block-matching module that enables large-scale UAV-satellite alignment in real time. These innovations allow GROMS to deliver up to 7.6× higher efficiency than commercial tools, while preserving accuracy and robustness across diverse terrains and environmental conditions.

It is worth emphasizing that our earlier work (Yao et al., 2024) primarily focused on UAV geolocalization in GNSS-denied environments via terrain-weighted constraint optimization. Extending this line of research, GROMS advances beyond geolocalization toward real-time orthophoto generation by incorporating terrain-constrained SLAM, along with innovative rectification and blending strategies, thereby achieving both computational efficiency and mapping accuracy.

3. Methodology

The workflow of GROMS is illustrated in Fig. 1. The first module selects UAV keyframes, partitions georeferenced satellite imagery into blocks, dynamically manages and merges these blocks, and performs both UAV-UAV and UAV-satellite image matching. As orthophoto generation requires accurate camera poses for ortho-rectification, the second module introduces a GPS- and IMU-free method for absolute UAV pose estimation and optimization. In this module, georeferenced satellite imagery and DEMs are used only as map priors to derive virtual control points (VCPs), which impose terrain constraints within a

Table 1
Keyframe selection parameters used in GROMS.

Parameter	Symbol/Setting	Value
Minimum image overlap	–	70%
Rotation threshold	–	45°
Candidate period	Δt	0.5–1.0 s
Fixed-interval variants	–	15–30 frames

monocular SLAM framework. These priors do not provide direct GNSS or inertial measurements but serve to constrain pose estimation and sliding-window optimization, enabling accurate and drift-reduced pose computation in GPS-free environments. The third module proposes a hybrid ortho-rectification method for UAV imagery, combined with an incremental multi-band blending strategy to achieve seamless and real-time orthophoto generation. The outputs of GROMS include the final georeferenced orthophoto, the estimated UAV poses in an absolute coordinate frame, and optionally the reconstructed 3D map points.

3.1. Satellite image management and matching

In GPS-free scenarios, GROMS estimates UAV image poses by matching them against satellite imagery. To improve matching efficiency and accuracy, given the typically large coverage of satellite imagery, the system divides satellite images into 128×128 pixels tiles, as shown in Fig. 2. This tile size is selected as a trade-off between computational efficiency and matching robustness. Smaller tiles increase the number of tiles to be loaded and merged, leading to higher I/O overhead and reduced spatial context, while larger tiles introduce excessive non-overlapping regions relative to the UAV image, increasing memory consumption and degrading matching performance. In addition, the chosen size is compatible with common learning-based matching frameworks that require input dimensions to be multiples of 8. The satellite image tiles within the UAV image's corresponding range R are then merged for subsequent matching.

Keyframe selection must balance computational efficiency and geometric reliability for SLAM-based pose estimation and orthophoto generation. GROMS follows the keyframe selection philosophy of ORB-SLAM2 (Mur-Artal and Tardós, 2017), while adapting it to the requirements of large-scale orthophoto mapping. Processing every frame in high-frame-rate UAV videos introduces significant redundancy due to extremely small inter-frame baselines, whereas overly sparse keyframes reduce image overlap and degrade matching robustness. GROMS therefore adopts a hybrid strategy combining fixed-interval candidate sampling with adaptive geometric constraints. The spatial distance between sampled frames is defined as

$$\Delta s = v \cdot \Delta t \quad (1)$$

where v denotes UAV flight speed, and Δt is the temporal sampling interval. For high-frame-rate UAV videos, GROMS extracts candidate frames at a fixed temporal interval corresponding to half of the video frame rate. For example, when the video frame rate is 30 fps, and the UAV flies at approximately 10 m/s, candidate frames are sampled every 15 frames, yielding $\Delta t = 0.5$ and a spatial spacing of $\Delta s \approx 5$ m.

As shown in Table 1, candidate frames are extracted at a fixed interval (typically half of the video frame rate) and promoted to keyframes only if predefined overlap and rotation constraints are violated. This strategy reduces redundant computation while preserving a sufficient baseline for robust triangulation and stable orthophoto generation.

Orthophoto generation relies on accurate camera pose, which requires precise feature matching. Differences in viewpoint, illumination, season, and radiometric between UAV and satellite images limit the accuracy and robustness of traditional matching methods. Therefore, SuperPoint (DeTone et al., 2018) is employed to extract local features and compute descriptors from the UAV keyframe and satellite blocks; MINIMA (Ren et al., 2025) is then applied for feature matching,

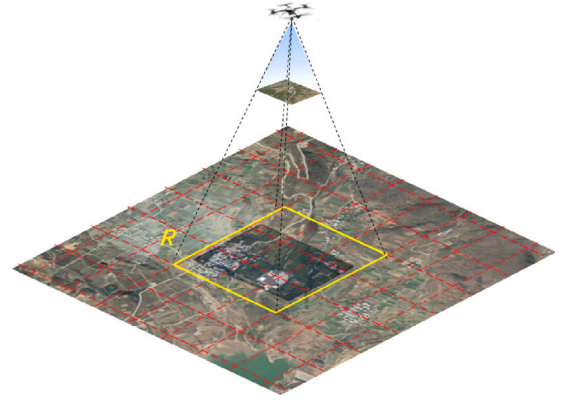


Fig. 2. Satellite image tiling and merging. Red dashed lines show tile boundaries; the yellow region R marks the UAV image's corresponding area.

and RANSAC removes mismatches to yield robust correspondences. Additionally, a keyframe pre-rotation strategy estimates the current frame's orientation using matched points and applies it to the next keyframe to handle large rotations. This improves feature matching under large rotations and gives GROMS rotation invariance. In contrast, the study (Kim et al., 2020) relies on IMU angle data for viewpoint estimation.

3.2. Absolute pose estimation and optimization of UAV images

Orthophoto generation needs absolute camera poses, but traditional SLAM provides only relative ones. Existing SfM and SLAM methods (Zhao et al., 2021; Hinzmann et al., 2018) rely on GPS and IMU to initialize camera poses. In GPS-free scenarios, initial UAV poses are estimated via a PnP algorithm using Virtual Control Points (VCPs), which are then integrated into SLAM for terrain-constrained sliding window optimization. This process yields georeferenced UAV poses suitable for accurate orthorectification.

As shown in Fig. 3, the system first extracts SuperPoint features from keyframes F_{n-1} , F_n , and the satellite image. Then, MINIMA is used to match F_{n-1} and F_n with the satellite image, resulting in pairs of matched features (A_i, C_k) and (B_j, C_k) . If A_i and B_j also form a matched pair via MINIMA, then A_i , B_j , and C_k constitute a mutually consistent and reliable feature triplet. The system thus treats the satellite-derived feature point C_k as a VCP. The 2D coordinates (x_k, y_k) of point C_k on the satellite image are pre-determined. The elevation value z_k of C_k is acquired through cubic spline interpolation over the DEM corresponding to the satellite image. Once the 3D Virtual Control Point (3DVCPs) C_k and its corresponding feature points (A_i, B_j) are established, the PnP algorithm is applied to estimate the initial absolute pose of the UAV image.

Seasonal, lighting, and viewpoint differences between UAV and satellite images can cause matching errors, and texture-sparse areas may lack enough 3DVCPs to solve the PnP problem. Moreover, frame-by-frame PnP estimation overlooks the geometric co-visibility between adjacent frames, leading to large pose fluctuations that are unsuitable for subsequent orthophoto correction. To address this issue, GROMS triangulates co-visible features (E_m, F_n) after PnP-based pose estimation of F_n and F_{n+1} , generating 3D map points (Green triangles D_i in Fig. 3). Meanwhile, to reduce pose drift from SLAM errors, 3DVCPs are integrated into SLAM for terrain-constrained sliding window optimization.

Visual SLAM typically employs Bundle Adjustment (BA) for pose optimization.

$$e = \min_{\xi, P} \sum_{i=1}^m \sum_{j=1}^n \left\| r(\xi_i, P_j) - p_{ij} \right\|^2 \quad (2)$$

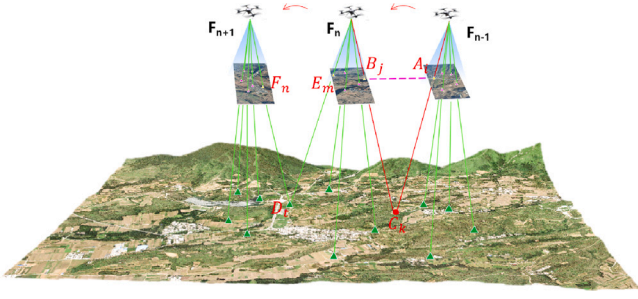


Fig. 3. Virtual Control Point Extraction. Green triangles (D_i) are triangulated map points; red circles (C_k) are 3D virtual control points validated by matching.

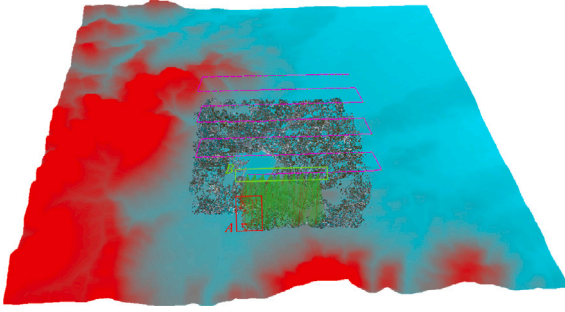


Fig. 4. UAV image absolute pose estimation and terrain-constrained sliding window optimization using 3DVCPs. Blue-to-red gradient indicates elevation. Colored points in box A are triangulated map points; red points are virtual control points. Green region B shows the system's sliding window optimization scope.

Where $r()$ denotes the reprojection function, i represents the index of the i -th image, j indicates the j -th map point observed in the image. Bundle Adjustment refines the initial PnP-estimated pose ξ and the map point P (such as D_1 in Fig. 3) by iteratively minimizing the reprojection error between P and the observed feature locations p (such as E_m and F_n in Fig. 3), which serves as the odometry constraint during pose optimization. Nevertheless, this approach yields only relative UAV poses, and as time progresses, accumulated errors in the estimated poses lead to noticeable drift. To mitigate this problem, we also project the extracted 3DVCPs C_k and use the difference between their projections and observed positions (A_i, B_j) as terrain constraints. The error term e can then be expressed as:

$$e = \min_{\xi, P} \sum_{i=1}^m \sum_{j=1}^n \left\{ \rho \left(\left\| r(\xi_i, P_j) - p_{ij} \right\|^2 \right) + \alpha_j \times \left\| r(\xi_i, C_j) - p_{ij} \right\|^2 \right\} \quad (3)$$

Where ρ denotes the Huber kernel function, which is primarily used to reduce the influence of large reprojection outliers. C represents the virtual control point. The coefficient α is defined as follows:

$$\alpha_j = \begin{cases} 1, & \text{otherwise.} \\ \frac{1}{\left\| Z_j - z_j \right\|}, & \text{if } \left\| Z_j - z_j \right\| \geq 3 \end{cases} \quad (4)$$

Where Z_j denotes the elevation of the virtual control point and z_j represents the elevation of the corresponding map point. The coefficient dynamically adjusts the weight of the terrain constraint during BA optimization. As shown in Fig. 4, combining odometry with terrain constraints reduces error accumulation, suppresses trajectory drift, improves pose accuracy, and enables relocalization without loop closure.

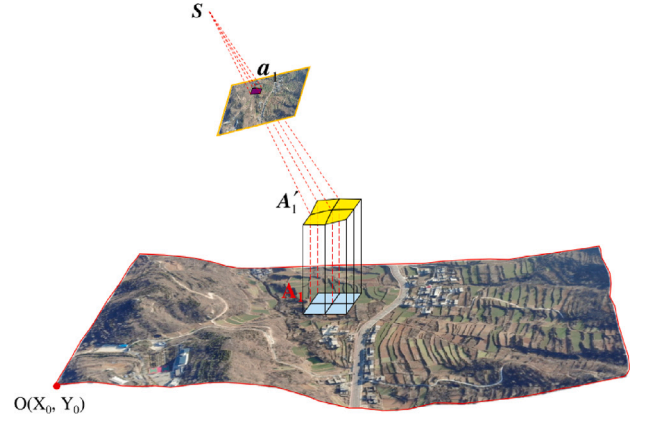


Fig. 5. The process of indirect orthorectification.

To enhance both efficiency and accuracy, the system employs a sliding window optimization strategy, as shown in Fig. 4, where only keyframes within a local temporal window are jointly refined using BA. This reduces computational overhead while smoothing the UAV's estimated trajectory. In texture-deficient regions, where reliable matching fails to yield enough 3DVCPs for pose estimation, the system applies a constant-velocity model that extrapolates the current frame's pose Rt_n based on the previous pose Rt_{n-1} and Rt_{n-2} .

$$Rt_n = Rt_{n-1} \cdot (Rt_{n-1} \cdot Rt_{n-2}^{-1}) \quad (5)$$

In our framework, satellite imagery and DEMs serve as auxiliary priors rather than direct sources for orthophoto generation. They provide a georeferenced global reference frame that constrains UAV pose estimation, which is crucial in GPS-free environments. Without satellite/DEM priors, UAV-only methods would accumulate errors over long flights and lack absolute georeferencing, rendering real-time orthophoto mapping infeasible. Therefore, integrating UAV imagery with satellite/DEM priors is essential for achieving high-resolution, drift-free, and real-time orthophoto mapping in GPS-free conditions.

3.3. Real-time UAV image orthorectification and fusion

The goal of orthorectification is to use the estimated image pose and a surface model to correct perspective distortions in aerial imagery caused by viewing angle and surface geometry. Therefore, GROMS first estimates and optimizes pose (R, t) using terrain-constrained monocular SLAM, then applies DEM-based pixel- and block-level correction for efficient, real-time orthorectification. To reduce color discrepancies and seams caused by lighting, exposure, GROMS employs multi-band blending to incrementally fuse images, producing seamless, consistent results in overlaps.

As illustrated in Fig. 5, indirect orthorectification starts from a pixel A_1 in the orthophoto, back-projects it to obtain its corresponding coordinate a_1 in the original image, and then assigns the grayscale value of a_1 to A_1 . The coordinate of A_1 in the orthophoto is (X', Y') and its corresponding geographic coordinate $A'_1(X, Y)$ is calculated based on the lower-left corner $O(X_0, Y_0)$ and the map scale M . The ground elevation Z corresponding to A'_1 is interpolated from the DEM using bicubic interpolation. Using the UAV image's pose (R, t), and camera intrinsic param, the 3D point $A'_1(X, Y, Z)$ is projected to its image coordinate $a_1(x, y)$ via the collinear equation model. The grayscale value at a_1 is then interpolated and assigned to the pixel A_1 in the orthophoto.

$$\begin{cases} x - x_0 = -f \frac{a_1(X - X_S) + b_1(Y - Y_S) + c_1(Z - Z_S)}{a_3(X - X_S) + b_3(Y - Y_S) + c_3(Z - Z_S)} \\ y - y_0 = -f \frac{a_2(X - X_S) + b_2(Y - Y_S) + c_2(Z - Z_S)}{a_3(X - X_S) + b_3(Y - Y_S) + c_3(Z - Z_S)} \end{cases} \quad (6)$$

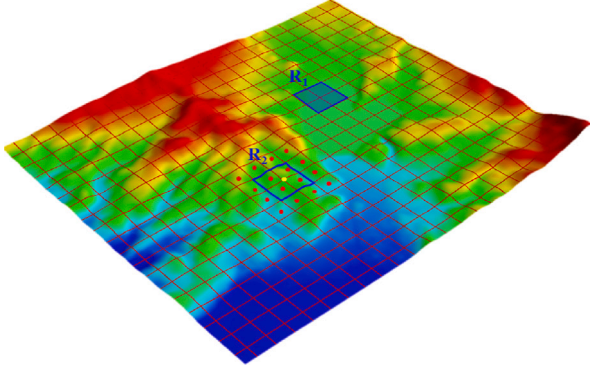


Fig. 6. The pixel-wise and block-wise orthorectification of the UAV imagery based on DEM elevation data.

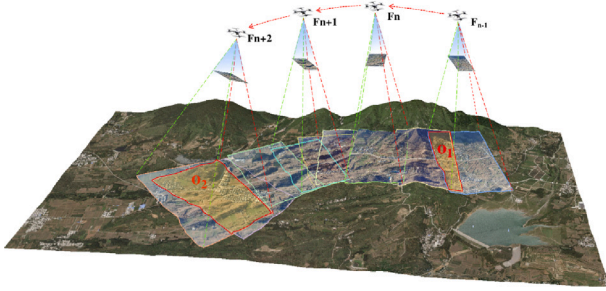


Fig. 7. The incremental fusion of orthorectified images using the multi-band blending strategy.

Where (X_0, Y_0, f) represents the camera intrinsic param, while, (a_i, b_i, c_i) and (X_S, Y_S, Z_S) denote the image pose (R, t) . By performing the above operations sequentially for each pixel, the corrected orthophoto can be obtained.

While the colinear model produces accurate orthophotos, per-point back-projection is computationally expensive, limiting real-time use for large, high-resolution images. To address this issue, GROMS adopts a hybrid strategy combining pixel-wise and block-wise correction. It first partitions the orthophoto into virtual blocks and calculates the standard deviation of the DEM elevation values within each block. In flat areas like region R_1 (Fig. 6), it uses the block's average DEM to skip per-point interpolation, then applies a remapping function to correct the tile, thereby enhancing processing speed. In contrast, in rugged areas like region R_2 , it uses pixel-wise correction with per-point interpolation to ensure accurate rectification and avoid visible distortions in the orthophoto. The remapping function is defined as:

$$O(X', \cdot Y') = I(Re(X, \cdot Y)) \quad (7)$$

where $O()$ denotes the orthorectified image, $I()$ represents the UAV image, and $Re(X, Y)$ is the mapping function applied to the coordinates (X, Y) . After the mapping, the system applies Lanczos interpolation using an 8×8 neighborhood around (X, Y) to perform convolution-based resampling, thereby achieving high-quality image output.

Variations in illumination, sensor radiometry, terrain, and geometric correction errors often introduce visible seams, color inconsistencies, and misalignments in overlapping orthorectified images. To reduce these artifacts, GROMS performs incremental blending on the corrected imagery using a multi-band blending approach, as illustrated in overlapping regions O1 and O2 in Fig. 7. This method utilizes multi-resolution decomposition to blend images across different frequency bands, ensuring smooth transitions in low-frequency regions while preserving high-frequency details. The process begins by decomposing images into a Laplacian pyramid, which separates the images into

multiple frequency bands. For each image, the Laplacian pyramid at level l is computed as:

$$L^l(I) = G^l(I) - G^{l+1}(I) \uparrow \quad (8)$$

Where $G^l(I)$ is the Gaussian pyramid of image I at level l , and $\uparrow$ denotes upsampling to match the resolution of $G^l(I)$. The Gaussian pyramid is constructed by iteratively applying Gaussian blur and downsampling:

$$G^{l+1}(I) = \text{downsample}(\text{blur}(G^l(I))) \quad (9)$$

For images A and B, at each pyramid level l , the blended Laplacian pyramid is computed as:

$$L_{\text{blend}}^l = W^l \cdot L^l(A) + (1 - W^l) \cdot L^l(B) \quad (10)$$

Where W^l is the weight map at level l , typically obtained by downsampling the original weight map to match the resolution of level l . $L^l(A)$ and $L^l(B)$ are denoted as Laplacian pyramids of A and B, respectively. This ensures that blending weights are consistent across scales.

The final blended image is reconstructed by collapsing the blended Laplacian pyramid, starting from the coarsest level L_{blend}^N and iteratively upsampling and adding finer details:

$$I_{\text{blend}} = L_{\text{blend}}^N \uparrow + L_{\text{blend}}^{N-1} \uparrow + \dots + L_{\text{blend}}^0 \quad (11)$$

This approach smooths low-frequency regions while preserving high-frequency details like building edges and roads, producing natural, seamless overlaps ideal for real-time orthophoto fusion.

4. Experiments

This section first introduces the dataset and evaluation metrics, followed by comprehensive experiments comparing the results of GROMS with open-source programs and state-of-the-art commercial software, along with ablation studies to evaluate its real-time performance, accuracy, and robustness.

4.1. Dataset and metric analysis

The NUP DroneMap dataset, developed by Bu et al. (2016), was collected using rotary-wing UAVs flying at various altitudes and over diverse scenes. It primarily contains keyframe images, trajectory data, and camera metadata. Detailed specifications are presented in Table 2(a–f). To address the limitations of the NUP DroneMap dataset in altitude, flight paths, and viewing angles, we collected additional data using multiple UAV platforms across plains, hilly terrains, urban, and rural areas. These datasets cover varied terrain, altitudes, oblique views, complex paths, and lighting, enabling thorough evaluation of GROMS's performance in pose estimation and real-time orthorectification. Details of the 14 newly collected UAV datasets, referred to as the XD Lab DroneMap, are listed in Table 2(g–t). In GNSS-free scenarios, the closer the acquisition time and resolution of satellite images are to those of UAV images, the fewer feature changes occur, leading to improved matching performance with MINIMA as well as higher accuracy and robustness in UAV positioning and orthomapping. Google satellite imagery at zoom levels 16 to 19 and the Advanced Land Observing Satellite (ALOS) DEM with a resolution of 12.5 m were employed as auxiliary inputs to support UAV pose estimation.

The GPS trajectory recorded by the UAV serves as ground truth for calculating the Root Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) in the x, y, and z directions across different positioning methods. MAE reflects average error magnitude, with smaller values indicating higher accuracy, while RMSE highlights outliers, where lower values imply greater stability. Additionally, manually surveyed GCPs are also used as ground truth to assess the xy-plane geolocation accuracy of orthophotos using different methods.

Table 2
Characteristics of the UAV datasets (ED: Elevation Difference, KFs: Key Frames).

Set	Location	Scene	Resolution	KFs	Camera	ED (m)	Alt. (m)	Pitch (°)	Area (km ²)	Flight date
a	Shenzhen	Towns	1920*1080	835	Phantom3	31.7	161.8	-90	0.606	Feb. 2016
b	Luoyang	Towns	1920*1080	402	Phantom3	42.3	198.7	-90	0.912	Apr. 2016
c	Hanzhong	Village	1920*1080	415	Phantom3	11.2	258.3	-90	1.457	Mar. 2016
d	Hanzhong	Village	1920*1080	285	Phantom3	20.3	165.6	-90	0.526	Mar. 2016
e	Hengdong	Village	1920*1080	221	Phantom3	146.2	358	-90	0.936	Feb. 2016
f	Xi'an	City	1920*1080	457	Phantom3	23.4	254.5	-90	1.598	Jan. 2016
g	Zhaoyuan	Village	1920*1080	196	DJI H20t	9.5	199~258	-90.2	0.405	Jul. 2022
h	Zhaoyuan	Village	1920*1080	298	DJI H20t	9.5	500	-90	1.512	Jul. 2022
i	Dengfeng	Hilly	1920*1080	142	DJI P4R	25.3	500	-89.9	1.687	Mar. 2022
j	Dengfeng	Mountain	1920*1080	410	DJI H20t	180.8	753~1048	-91.3~-64.8	13.444	Jan. 2024
k	Dengfeng	Towns	1920*1080	514	DJI H20t	88.9	500	-89.7	8.089	Mar. 2023
l	Dengfeng	Mountain	1920*1080	395	DJI H20t	201.5	500	-89.8	5.727	Mar. 2023
m	Dengfeng	Mountain	1920*1080	225	DJI H20t	335	1030	-109.4~-69.9	9.45	Jan. 2024
n	Zhengzhou	City	1920*1080	340	DJI H20t	12.1	500	-89.9	2.022	May. 2023
o	Zhengzhou	City	1920*1080	323	DJI H20t	13.2	500	-89.8~-83.6	1.775	Dec. 2023
p	Dengfeng	Villages	640*512	126	DJI H20t	85.3	500	-89.6	1.307	Jan. 2021
q	Dengfeng	Hilly	640*512	271	DJI H20t	50.6	500	-90.4	0.945	Jan. 2021
r	Dengfeng	Hilly	7952*5304	191	RX1	339.2	800	-90	14.785	Jul. 2022
s	Dengfeng	Village	5280*3956	56	DJI M3E	48.6	660	-90	0.420	Apr. 2024
t	Zhengzhou	Village	5280*3956	57	DJI M3E	41.1	500	-90	3.404	Dec. 2023

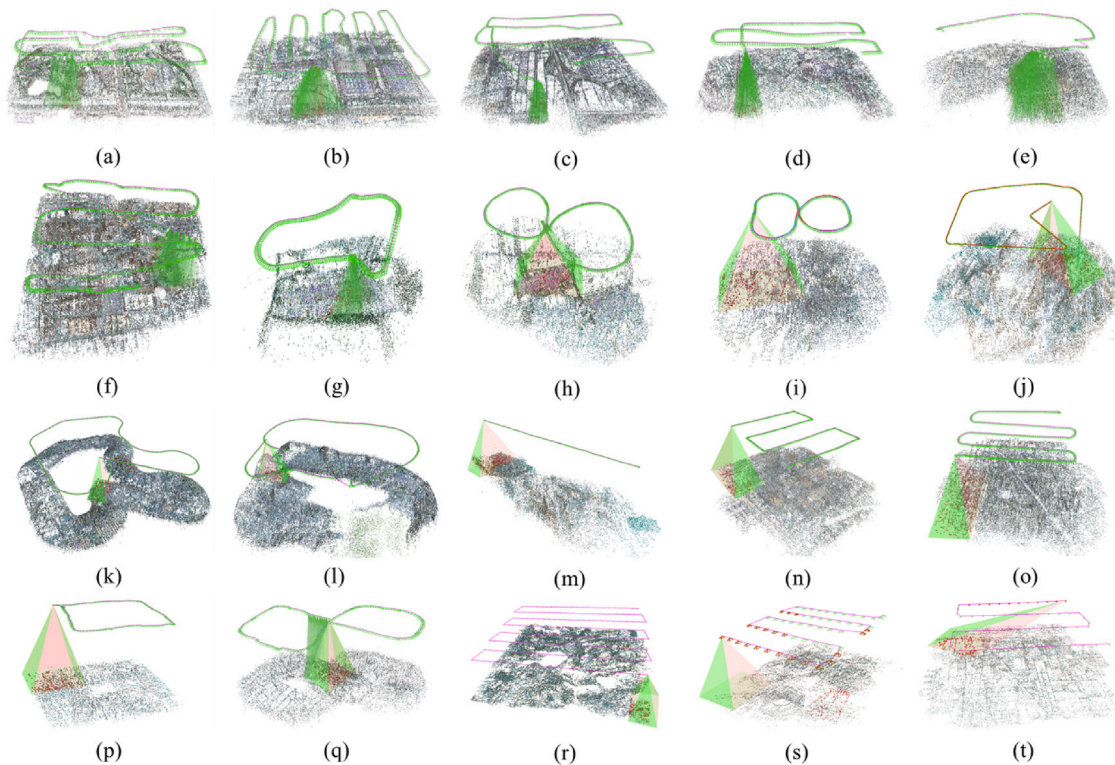


Fig. 8. Absolute localization results for 20 UAV datasets using GROMS under GPS-free conditions. Blue lines indicate the ground-truth trajectories, while red lines represent the trajectories estimated by GROMS. Colored points denote 3D map points generated by the system.

4.2. Robustness evaluation of GPS-free UAV pose estimation and orthophoto generation

Figs. 8 and 9 present the localization trajectories and orthophotos for 20 datasets from NUP DroneMap (a–f) and XDLab DroneMap (g–t). Results show GROMS robustly estimates UAV poses and generates orthophotos under varying paths, altitudes, angles, scenes, and seasons—without GPS.

As shown in Table 2 and Fig. 8, NUP DroneMap data were acquired using regular trajectories, low fixed altitudes, and a nadir-viewing camera. Terrain variation was minimal, and the spatial coverage was limited. To enhance diversity, XDLab DroneMap includes standard and complex trajectories in sets g to l. The UAVs typically operated at

altitudes exceeding 500 m. In sets g and j, variable-altitude flights were employed instead of fixed-altitude operations. During data acquisition for sets j, m, and o, the camera pitch was dynamically adjusted to introduce intentional obliqueness, rather than maintaining a strictly nadir view as in [Li et al. \(2023\)](#). Moreover, the flight areas covered broader regions, with set r spanning 14.7 km². Data covered plains, hills, urban, and rural areas, with elevation differences reaching up to 339.2 m. As shown in [Fig. 9](#) (sets g, h, k, l), UAVs traversed texture-sparse areas like cornfields, bare terrain, and reservoirs during data acquisition. In sets s and t, snow-covered imagery was captured during winter. Sets p and q collected thermal infrared data to evaluate nighttime localization and orthophoto generation. The combination of irregular trajectories, variable altitudes, oblique camera angles, diverse



Fig. 9. Thumbnail orthorectification results for 20 UAV image datasets. Sets p and q display thermal infrared orthophotos, whereas sets s and t correspond to orthophotos acquired under snow-covered winter conditions.

scene types, texture-sparse regions, and seasonal illumination variations presented significant challenges to the GROMS system in GPS-free pose estimation and orthophoto generation, thereby highlighting its robustness.

4.3. Visual comparisons with state-of-the-art methods

An ideal orthophoto should feature high spatial resolution, uniform visual appearance, minimal geometric distortion, and limited artifacts such as displacement, blur, and warping. To qualitatively assess orthophoto quality, we visually compared the results produced by Map2DFusion, three state-of-the-art commercial software—Photoscan, Pix4Dmapper, and ContextCapture (Bentley Systems, 2025)—and our GROMS system, using both the public NUP DroneMap and our custom datasets.

As illustrated in Fig. 10, orthophotos of water bodies, buildings, roads, vegetation, and village areas were generated from the NUP DroneMap dataset. Map2DFusion fits a terrain plane using sparse map points, which improves processing efficiency but fails to model non-planar environments effectively. Its simple homography-based projection causes mismatches in regions with significant elevation variation, thereby degrading orthophoto quality. In Fig. 10, Map2DFusion orthophotos exhibit misalignments near buildings and roads, ghosting in water and vegetation, and structural distortions in villages. In the building regions shown in Figs. 10(b) and (e), projection discrepancies cause all three commercial software and GROMS to produce visibly tilted buildings. Because the generated DSMs struggle to capture detailed 3D structures of buildings, the orthophotos produced through digital differential correction show various artifacts: ContextCapture yields irregular

building boundaries, Photoscan introduces distortions, and Pix4Dmapper suffers from edge misalignments. In the road and vegetation regions (Figs. 10(c) and (d)), low DSM fidelity and pose errors cause twisting and misalignment in orthophotos from the three commercial tools. In contrast, GROMS generates smoother, more natural imagery with fewer outliers and distortions.

Map2DFusion requires accurate pose and terrain fitting parameters for orthophoto generation, which are absent in the XDLab DroneMap dataset. Fig. 11 compares orthophotos generated by GROMS and three commercial software across diverse representative scenes, including water, highways, rural and urban areas, and mountain regions. As shown in Fig. 11(a), in texture-sparse regions such as water bodies, ContextCapture fails to accurately reconstruct, resulting in visible holes in the orthophotos. Photoscan and GROMS exhibit suboptimal blending over water areas, producing visible bright spots in the orthophotos. Fig. 11(b) illustrates highway distortions in all commercial software due to DSM inaccuracies, whereas GROMS maintains visually straight roads without apparent warping. In rural areas with dense low-rise buildings (Fig. 11(c)), ContextCapture and Pix4Dmapper struggle to extract structural details from point clouds, leading to edge deformation in orthophotos. Photoscan shows large missing areas, while GROMS better preserves building features with minimal deformation. In the tall building area (Fig. 11(e)), buildings generated by GROMS appear tilted due to DEM limitations, yet structural details are preserved. In contrast, outputs from commercial software exhibit evident loss and discontinuity at building edges and corners. In mountainous and hilly areas (Figs. 11(d) and (f)), GROMS and commercial software reconstruct terrain reasonably well, with no misalignment or deformation observed



Fig. 10. Comparison of orthophotos over water bodies, buildings, roads, vegetation, and village areas from the NUP DroneMap dataset using different methods.

along the mountaintop path in Fig. 11(d). However, Photoscan shows holes and blurring in the snow-covered area (Fig. 11(f)) due to low overlap. Pix4dMapper fails to estimate thermal image poses, preventing orthophoto generation (Fig. 11(g)).

Fig. 12 compares the orthoimage performed by Photoscan, GROMS, and a stitching method. Both GROMS and Photoscan employ digital differential correction based on terrain data, producing image deformations that accurately represent terrain variations. As shown in Figs. 15(c) and (d), GROMS orthophotos closely match Google Earth's 3D terrain. The stitching method, relying on parametric models such as homography (Kim et al., 2020; Ren et al., 2022), cannot handle perspective distortions caused by terrain undulations and viewpoint variations. Although geometrically regular, the corrected images exhibit significant errors and are unsuitable for mosaicking in non-planar terrain.

Fig. 13 illustrates GROMS correcting oblique UAV images captured in mountainous regions. Despite significant tilt and rotation, GROMS achieves accurate orthorectification through pose estimation combined with DEM-based differential correction. In contrast, study (Li et al., 2023) requires a nadir-facing camera to reduce tilt-induced distortion during orthophoto generation.

4.4. Accuracy evaluation

To effectively evaluate GROMS-derived DOMs accuracy, we use GCPs measured by GPS for absolute error assessment. As shown in Fig. 14, for set r, 16 GCPs were manually surveyed. The error is defined as the Euclidean distance between each marker (center or corner) and its corresponding location on the orthophoto.

Table 3 reports horizontal errors Δx , Δy , and median error Δxy for points on DOMs from three commercial tools and GROMS. It can be observed that among the three SfM-based tools, Photoscan demonstrates the highest orthophoto accuracy, followed by ContextCapture, with Pix4dMapper lowest. All three achieve sub-meter accuracy. GROMS yields an MAE of 1.11 m in x and 0.64 m in y. Although GROMS exhibits the largest x-error, its y-accuracy exceeds that of ContextCapture and Pix4dMapper. Notably, GROMS is GPS-free, using only visual SLAM and terrain-constrained BA for orthophoto generation. In contrast, commercial software requires accurate GPS and global BA to minimize localization drift and ensure reconstruction scale and geo-referencing. In Figs. 15(a) and (b), the GROMS orthophoto is overlaid and alternately compared with satellite imagery, revealing no significant feature misalignment. In Figs. 15(c) and (d), the GROMS orthophoto aligns precisely with the Google Earth 3D terrain. Compared to commercial software, although GROMS exhibits slightly lower



Fig. 11. Orthophotos from the XDLab DroneMap dataset generated by three commercial software packages and GROMS. Set f contains snow-covered imagery, while set g presents thermal infrared orthophotos.

Table 3

GCP errors of different methods. The unit is meters.

GCP	GROMS (Ours)			Photoscan			Pix4DMapper			ContextCapture		
	Δx	Δy	Δxy	Δx	Δy	Δxy	Δx	Δy	Δxy	Δx	Δy	Δxy
C1	-1.500	0.234	1.519	-0.131	-0.757	0.769	-1.318	-0.820	1.552	1.322	0.883	1.590
C2	-0.462	-0.253	0.527	0.550	0.033	0.551	-0.118	-0.537	0.550	0.245	0.450	0.512
C3	0.794	-0.215	0.823	0.731	-0.185	0.754	0.997	-0.296	1.040	-0.740	-0.482	0.883
C4	-1.055	0.617	1.222	0.871	0.688	1.110	1.073	1.772	2.072	-0.783	-1.961	2.111
C5	-0.355	0.806	0.881	-0.689	1.110	1.306	-1.601	1.910	2.492	1.205	-1.470	1.901
C6	-0.813	0.740	1.099	-0.549	0.649	0.850	-1.340	0.902	1.615	1.371	0.077	1.373
C7	-3.277	1.047	3.440	0.286	-0.364	0.463	0.280	-0.949	0.989	0.094	0.730	0.736
C8	-0.387	-0.318	0.500	0.617	0.300	0.686	0.890	0.655	1.105	-0.577	-0.928	1.093
C9	-0.728	1.385	1.565	-0.404	0.230	0.464	-0.748	0.493	0.896	0.726	-0.509	0.886
C10	-0.603	0.273	0.662	-0.093	-0.296	0.310	-1.028	-0.355	1.088	0.843	0.518	0.990
C11	0.949	0.362	1.016	0.270	0.036	0.273	0.564	-0.144	0.582	-0.191	-0.202	0.278
C12	1.012	1.210	1.577	-0.001	0.957	0.957	-0.467	2.021	2.074	0.583	-1.701	1.798
C13	2.988	-1.414	3.306	0.070	0.967	0.970	-0.599	2.072	2.157	0.514	-1.063	1.180
C14	-0.295	1.443	1.473	-0.530	0.665	0.850	-1.409	1.342	1.946	0.841	-1.202	1.467
C15	2.308	1.018	2.522	0.340	-0.095	0.353	0.515	-0.198	0.552	-0.469	-0.131	0.487
C16	0.514	-0.200	0.552	0.477	-0.401	0.623	0.737	-1.213	1.420	0.240	1.104	1.130
MAE	1.111	0.640		0.414	0.444		0.819	0.958		0.568	0.761	

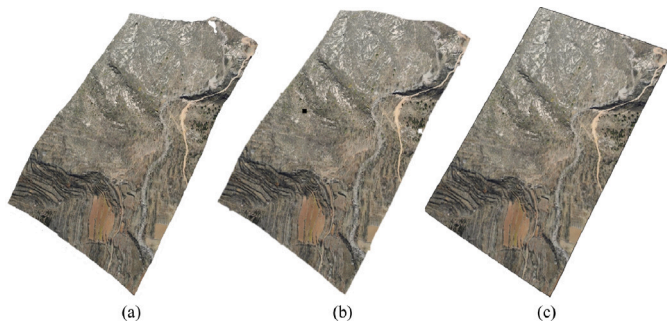


Fig. 12. Comparison of single-image correction results in areas with large elevation changes.

quantitative accuracy than SfM-based methods, its orthophotos' visual quality is comparable, and its real-time precision satisfies most practical requirements.

4.5. Efficiency evaluation

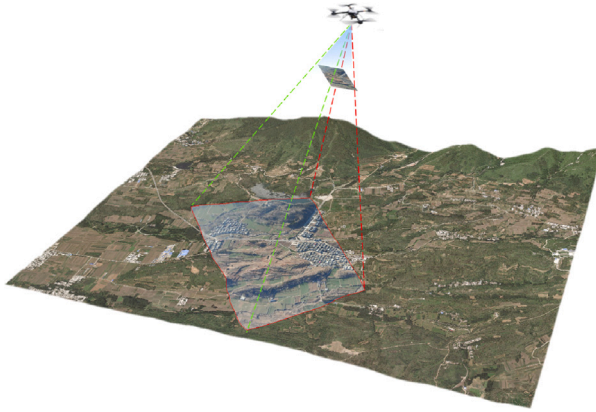
GROMS and Map2DFusion were executed on a 64-bit Ubuntu system, while the remaining commercial software was evaluated on 64-bit Windows 10, following their officially recommended platforms. All experiments were conducted on the same laptop equipped with an Intel i7-10750H CPU (6 cores @ 2.6 GHz), an NVIDIA GTX 1660 Ti Mobile GPU with 6 GB VRAM, 32 GB RAM, and a 1 TB NVMe SSD. All reported runtimes are wall-clock measurements.

In this work, "real-time" refers to the system's ability to sustain continuous online orthophoto generation with bounded end-to-end latency and stable throughput, rather than achieving a high frame rate

Table 4

Runtime comparison of different methods across two datasets. The unit is seconds.

Dataset	Pose estimation	Ortho-generate	Ablation study	Map2DFusion	Photoscan	Pix4dMapper	ContextCapture
NUP DroneMap	1838.487	966.903	2784.653	174.333	7057.5	10 200	11 520
XDLab DroneMap	3217.309	664.078	1994.175	–	10 908	19 050	39 420

**Fig. 13.** GROMS single-image correction results for tilted UAV images in mountainous terrain.**Fig. 14.** Verification of GCP errors in dataset r. Absolute error is defined as the Euclidean distance between the marker center and the GCP location, indicated by red labels and measured in meters.**Table 5**

End-to-end throughput and latency statistics of GROMS under online streaming operation.

Metric	Value
Average input image resolution	20.7 MPix/frame
Average end-to-end latency	1.68 s/frame
Latency p50 (median)	1.62 s
Latency p90	2.33 s
Latency p99	2.77 s
Sustained throughput	12.4 MPix/s
Processing mode	Online streaming

Table 6

Module-level runtime breakdown of GROMS (ms)

Module	Mean	p50	p90	p99
Tile I/O & merge	0.8	0.8	1.6	1.9
Keyframe selection	18.8	16.8	22.3	30.7
Pre-rotation	1.1	1.1	1.7	2.1
SuperPoint feature extraction	103.2	100.9	170.8	190.5
MINIMA matching	135	120.4	162	260.7
RANSAC filtering	0.7	0.7	1.1	1.4
VCP extraction	18.5	18.3	28.8	30.2
Triangulation	25.2	24.7	30.3	38.1
EPnP	6.4	6.3	6.9	7.8
Sliding-window optimization	620.8	618.8	773.1	811.8
Pose estimation (total)	930.5	908.8	1198.6	1375.2
Orthorectification	540.8	520.3	820.2	1010.7
Multiband blending	210.3	197.1	310.2	388.6
Output writing	1.8	1.7	2.3	3
End-to-end per frame	1683.4	1627.9	2331.3	2777.5

alone. Accordingly, efficiency is evaluated using end-to-end processing time, throughput, and latency statistics at the system level. Table 4 summarizes the average runtime comparison (in seconds). Map2DFusion requires accurate precomputed keyframe poses and terrain fitting parameters, and therefore cannot process the XDLab DroneMap dataset. When such priors are available, Map2DFusion only performs orthorectification and image fusion on the NUP DroneMap dataset, which explains its low runtime in Table 4. In contrast, GROMS performs online pose estimation using visual SLAM without relying on GPS/IMU, and thus the reported runtime includes both pose estimation and orthophoto generation. The ablation study in Table 4 corresponds to GROMS generating orthophotos using a pixel-wise correction strategy.

On average, GROMS requires 0.93 s per frame for end-to-end pose estimation. This includes satellite tile I/O and merging, SuperPoint feature extraction, MINIMA feature matching, virtual control point (VCP) extraction, EPnP-based coarse pose initialization, 3D point triangulation and filtering, and sliding-window optimization. The EPnP module itself is computationally lightweight, taking less than 7 ms per keyframe on average. This is because GROMS extracts a small set of robust VCPs and employs the EPnP algorithm, which has linear computational complexity with respect to the number of points. Orthophoto generation requires an additional 0.75 s per frame, mainly consisting of differential orthorectification and multi-band blending. The complete pipeline, therefore, results in an average end-to-end latency of approximately 1.68 s per frame.

To characterize real-time performance beyond average runtime, we report end-to-end throughput and latency statistics on the stated hardware. Table 5 reports the sustained throughput of GROMS in terms of megapixels per second (MPix/s), together with latency percentiles (p50, p90, and p99) for the slowest pipeline stages. The results show that GROMS achieves stable system-level throughput while maintaining bounded end-to-end latency (p99 < 2.8 s), providing a quantitative characterization of real-time orthophoto generation beyond frame-rate-based evaluation. Furthermore, Table 6 provides a module-level runtime breakdown, reporting the mean, median (p50), p90, and p99 latency for tile I/O and merging, keyframe selection, pre-rotation, SuperPoint, MINIMA, RANSAC filtering, VCP extraction, EPnP, triangulation, sliding-window optimization, orthorectification, multiband blending, and output writing. Satellite imagery is pre-tiled and stored offline; thus, Tile I/O and merge reflect only online tile fetching and merging.

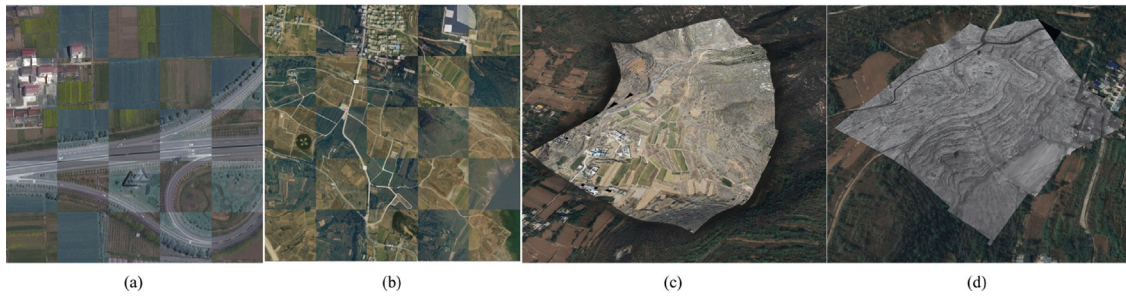


Fig. 15. Overlay of orthophotos generated by GROMS with satellite imagery and Google Earth 3D terrain.

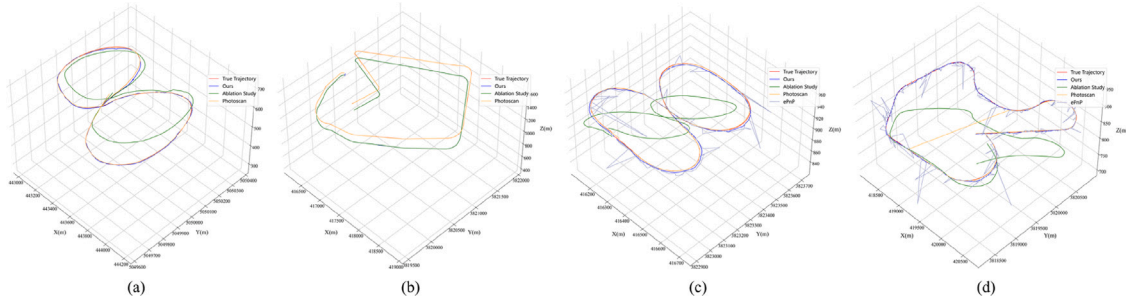


Fig. 16. Comparison of estimated positioning trajectories. Colored lines indicate the trajectories produced by different methods.

Table 7

MAE and RMSE results for localization of the XDLab DroneMap dataset.

	GROMS (ours)			Ablation study			PnP			Photocan		
	x (m)	y (m)	z (m)	x (m)	y (m)	z (m)	x (m)	y (m)	z (m)	x (m)	y (m)	z (m)
MAE	6.546	7.055	12.960	40.165	32.438	45.413	23.006	33.518	19.446	0.973	0.887	0.617
RMSE	10.327	9.829	18.129	44.838	35.616	51.085	46.794	63.607	36.186	1.146	1.089	0.728

SuperPoint feature extraction and MINIMA matching account for cross-view feature processing between UAV images and satellite tiles. Sliding-window optimization dominates the pose estimation stage, reflecting the computational cost of GNSS/IMU-free SLAM with map priors. This breakdown shows that EPnP is not a performance bottleneck, while feature extraction and matching, sliding-window optimization, and orthorectification dominate the overall latency.

Even when accounting for online pose estimation, GROMS significantly outperforms three representative commercial SfM-based software in terms of end-to-end efficiency. Specifically, GROMS achieves 2.6 $\times$, 4.3 $\times$, and 7.6 $\times$ speedups over Photocan, Pix4DMapper, and ContextCapture, respectively. The speedup is defined as the ratio of the total end-to-end wall-clock processing time per unit ground area, measured from raw UAV image ingestion to final orthophoto output, under identical output resolution and mapping coverage. All software is evaluated using identical orthophoto output resolution and comparable production settings following their standard orthophoto generation workflows. ContextCapture follows its standard workflow, including aerotriangulation, 3D reconstruction, and orthophoto production using photo position metadata, while GROMS performs online pose estimation via map-prior-assisted SLAM without GNSS/IMU and directly generates orthophotos in a streaming manner. Moreover, as the number of keyframes increases, the processing time of SfM-based commercial software grows rapidly due to their batch-oriented global optimization. In contrast, GROMS adopts SLAM-based incremental pose estimation and correction, resulting in approximately linear computational complexity with respect to the number of keyframes. This design enables GROMS to scale efficiently to large scenes without requiring all images to be available upfront or incurring prolonged offline processing.

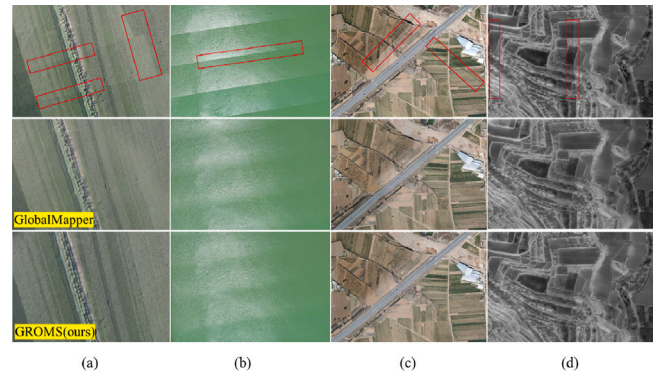


Fig. 17. Orthophoto blending results using GROMS's incremental multi-band blending and Global Mapper. Subfigure (d) shows the fused thermal infrared orthophoto.

4.6. Ablation study

UAV Pose Estimation and Optimization Using Terrain-Constrained Monocular SLAM. To evaluate terrain-constrained monocular SLAM for pose estimation, we compare GPS-based ground truth with the positioning results of GROMS, its ablation (without virtual control points), Photocan, and PnP method using the XDLab DroneMap dataset. In the ablation variant without virtual control points, the system still estimates absolute poses rather than relative ones. Specifically, after matching UAV images with satellite imagery, the world coordinates (X, Y) of matched features are obtained from the satellite image, and their elevations Z are interpolated from the DEM, yielding 3D points (X, Y,

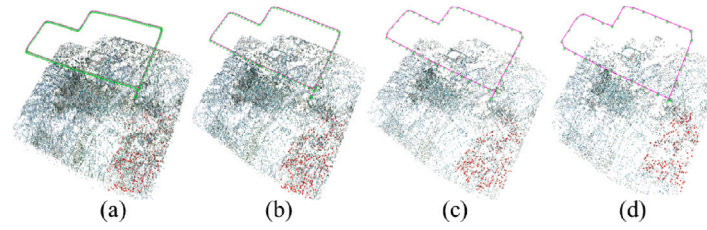


Fig. 18. The impact of extracting keyframes at fixed intervals (15, 30, 60, and 90 frames) on drone positioning accuracy and triangulated points.

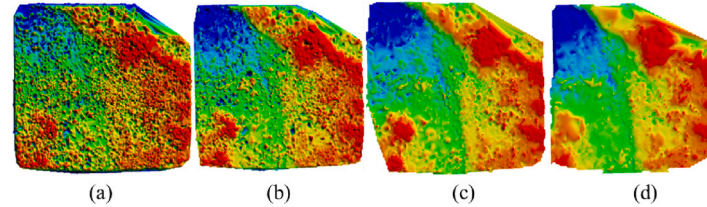


Fig. 19. Effect of frame spacing on triangulation quality and terrain smoothness. Delaunay triangulation is constructed from triangulated map points under different frame spacings (15, 30, 60, and 90 frames), illustrating the trade-off between point density and geometric accuracy.

Z). Together with the corresponding 2D pixels (x, y) in the UAV image, PnP is applied to estimate the absolute pose of each frame. Consecutive frames are then used for triangulation and bundle adjustment, which refines poses and 3D map points. Without VCP constraints, however, optimization relies only on geometric relations between UAV frames, leading to error accumulation, drift, and scale inconsistency. As shown in Table 7, Photoscan uses GPS as the initial value for global BA, yielding trajectories closely matching ground truth. GROMS achieves approximately 9 m level positioning accuracy using only visual data, without relying on GPS or other sensors. When virtual control point constraints are removed from GROMS (ablation study), positioning accuracy degrades to approximately 40 m. Although the PnP method achieves 25 m positioning accuracy, its high RMSE indicates significant trajectory error variation, resulting in an irregular and unsmooth path. As shown in Fig. 16, the trajectories estimated by GROMS and Photoscan are smooth and closely follow the GPS path. The ablation study, which does not incorporate virtual control point constraints, can still recover absolute coordinates; however, without VCPs as anchor constraints for terrain-constrained optimization, the system relies solely on geometric relations between UAV frames. Consequently, error accumulation and drift occur, and scale consistency is not well preserved. The PnP-estimated trajectory exhibits abrupt jumps, producing a discontinuous and non-smooth path. By leveraging SLAM-based pose estimation and virtual control point-constrained optimization, GROMS eliminates the need for GPS, IMU, and loop closure correction, while maintaining drift-free positioning and stable trajectories. This enables accurate pose estimation for reliable orthophoto generation.

Effect of Frame Spacing on Triangulation Quality and Matching Robustness. To analyze the impact of frame spacing on triangulation accuracy and matching robustness, we conducted a study using a 30 fps UAV video sequence, where keyframes were sampled at intervals of 15, 30, 60, and 90 frames.

As shown in Fig. 18, smaller frame intervals produce a larger number of triangulated map points, while increasing the interval significantly reduces point density. However, point quantity alone does not reflect geometric quality. Fig. 19 shows the Delaunay triangulations constructed from these map points to assess terrain consistency. When the frame spacing is small, the reconstructed terrain exhibits pronounced height fluctuations, indicating degraded triangulation accuracy due to insufficient baseline. Increasing the frame spacing yields smoother terrain surfaces with reduced height variation, better reflecting real-world topography.

Nevertheless, excessively large frame spacing leads to severe degradation in feature matching. As shown in Fig. 20, the number of matched

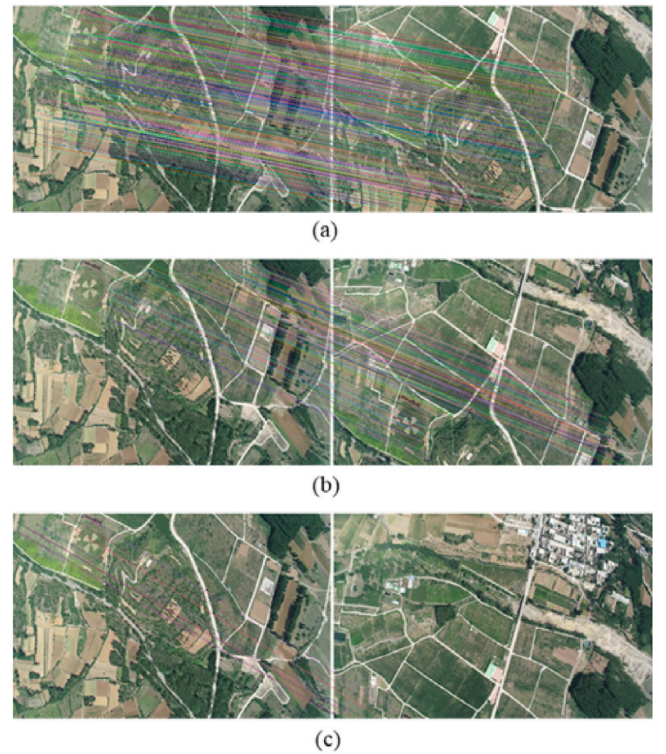


Fig. 20. Degradation of feature matching with increasing frame spacing. The number of matched feature points rapidly decreases as frame spacing increases, eventually preventing reliable VCP extraction and absolute localization.

feature points between the first frame and frames at large temporal distances drops rapidly (e.g., 452 matches at 120 frames, 136 at 240 frames, and only 12 at 300 frames). When reliable matching fails, VCP extraction becomes impossible, causing absolute localization and orthophoto generation to break down.

These results reveal a fundamental trade-off between triangulation accuracy and matching robustness. Based on this empirical analysis, GROMS adopts half-frame-rate candidate sampling as a practical compromise, providing a sufficient baseline for accurate triangulation while maintaining reliable matching and stable real-time performance.

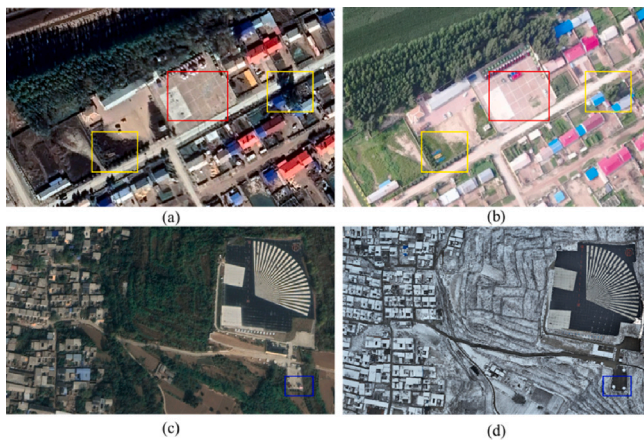


Fig. 21. Comparison between satellite basemaps and GROMS-generated UAV orthophotos.

Orthophoto Generation and Blending. To assess the effectiveness of GROMS's hybrid strategy combining per-pixel and block-wise correction, Table 4 presents the orthophoto generation times using per-pixel correction alone (the ablation study). On the NUP dataset, the hybrid method is 2.7 times faster than per-pixel correction, and on the XDLab dataset, it is approximately 3 times faster. As shown in the top row of Fig. 17, orthophotos exhibit visible color discrepancies and seams during mosaicking, primarily due to variations in lighting, sensor radiometric inconsistencies, and geometric correction errors. The second and third rows show the blending results of commercial software Global Mapper (Blue Marble Geographics, 2025) and GROMS. It is evident that our method effectively eliminates color discrepancies and seams between orthophotos, achieving smooth and natural transitions comparable to Global Mapper, thereby validating the effectiveness of GROMS's incremental multi-band blending strategy.

Comparison Between Satellite Basemaps and GROMS Orthophotos. To illustrate that UAV imagery is not redundant even with georeferenced satellite priors, Fig. 21 compares satellite basemaps and GROMS-generated UAV orthophotos over the same areas. Figs. 21(a) and 21(b) show a 0.5 m satellite image (October 2022) and a GROMS orthophoto generated from UAV imagery acquired in July 2022 at the same resolution. As highlighted in the red boxes, linear features such as road markings appear sharper and more continuous in the UAV orthophoto, indicating reduced edge width. In the yellow boxes, the UAV result exhibits a more natural color appearance with fewer shadow and occlusion artifacts. Figs. 21(c) and 21(d) further compare a 0.5 m satellite image (October 2022) with a 0.3 m GROMS orthophoto generated from UAV imagery collected in April 2024, demonstrating a higher effective GSD and finer structural details in the UAV result. The blue boxes highlight the improved temporal currency of UAV imagery. Overall, despite accurate georeferencing, satellite imagery is typically limited by lower spatial resolution and temporal update frequency, whereas GROMS-generated UAV orthophotos provide higher effective resolution, improved sharpness and linear-feature continuity, and better scene timeliness, confirming that UAV imagery remains essential rather than redundant.

Comparison Experiment with Gaussian Splatting. To further evaluate our method, we conducted a comparison with a representative learning-based approach, Gaussian Splatting (GS). As existing open-source GS/NeRF frameworks mainly focus on 3D reconstruction rather than orthophoto generation, and no publicly available implementations for orthophoto production are currently available, we employed ContextCapture with oblique images acquired by a five-lens camera to reconstruct a GS-based model in PLY format, from which orthophotos

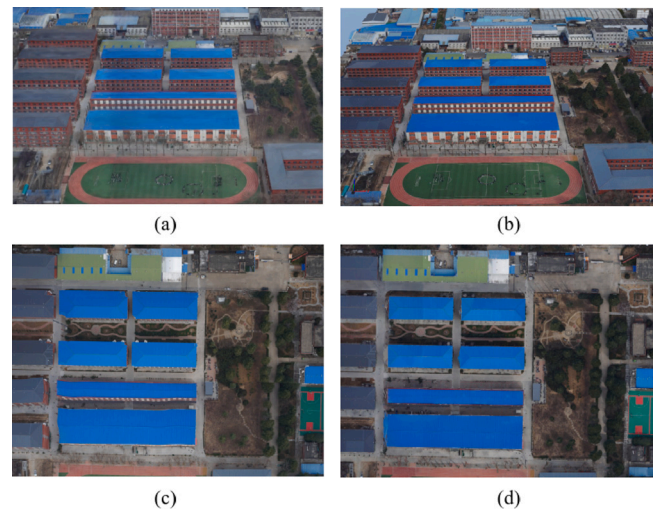


Fig. 22. Model and orthophoto comparison. (a) Gaussian Splatting-based 3D model; (b) Oblique photogrammetric 3D model; (c) Orthophoto generated by GROMS; (d) Orthophoto generated by the Gaussian Splatting model.

were subsequently generated. As illustrated in Fig. 22, GS produces visually appealing results with natural color reproduction, smooth shading, and fine structural details, for example, in vegetation and wires. It outperforms conventional oblique photogrammetry in terms of visual quality and reduces building deformation compared to GROMS in orthophotos. However, GS requires dense multi-view imagery, accurate initialization, and extensive offline processing, which limit its applicability in UAV scenarios with sparse overlaps, large viewpoint variations, and real-time requirements. In contrast, GROMS directly generates georeferenced orthophotos from UAV imagery in an online manner, while maintaining both efficiency and geometric accuracy.

5. Conclusions

This paper presents GROMS, a robust, accurate, and efficient real-time UAV orthophoto mapping system that operates without GPS or IMU measurements. GROMS integrates terrain-constrained monocular SLAM, hybrid ortho-rectification, and incremental multi-band blending, while leveraging georeferenced satellite imagery and DEMs solely as map priors to derive virtual control points and constrain pose estimation. Unlike existing methods that rely on GNSS/IMU measurements or ground control points, GROMS performs pose estimation and orthophoto generation in a GPS-free, map-prior-assisted manner, enabling high-precision pose estimation and real-time generation of seamless, color-consistent orthophotos. In terms of accuracy, GROMS achieves mean absolute errors of 1.11 m in the x-direction and 0.64 m in the y-direction under GPS-free conditions. In terms of efficiency, GROMS sustains stable system-level throughput with bounded end-to-end latency ($p99 < 2.8$ s) under continuous online processing, while achieving a $2.6\times$ to $7.6\times$ speedup over representative commercial software. Ablation studies confirm that VCP-SLAM achieves 9-meter-level localization accuracy and that hybrid rectification provides a $3\times$ speed improvement. The multi-band blending strategy ensures the seamless and color-consistent generation of orthophotos. In the future, we will focus on optimizing the computational efficiency of the SLAM module by integrating lightweight deep learning models for feature extraction and enhancing VCP selection to further improve pose estimation accuracy in feature-sparse environments, aiming to achieve sub-meter precision and faster real-time performance.

CRediT authorship contribution statement

Fushan Yao: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Chaozhen Lan:** Supervision, Resources, Methodology, Conceptualization. **Zijun Wei:** Writing – review & editing, Supervision, Formal analysis. **Tian Gao:** Writing – review & editing, Conceptualization. **Longhao Wang:** Writing – review & editing, Conceptualization. **Yiqiao Wang:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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